Integrated Sustainability Assessment of Municipal Solid Waste Management Systems: A Case Study of Marrakech

Material Flow Analysis (STAN) and Cost Efficiency Evaluation

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Abstract

This study assesses alternative municipal solid waste management systems in Marrakech, Morocco, by combining material flow analysis (MFA) and cost efficiency analysis, including three waste management scenarios to evaluate waste flows, recycling performance, composting potential, landfill diversion, resource recovery opportunities, and economic feasibility.

The results show that integrated waste management approaches that include source segregation, recycling, and composting could greatly enhance resource recovery and reduce reliance on landfill while supporting circular economy goals, and the findings can serve as guidelines for sustainable municipal waste management planning and decision-making.

Keywords: Material Flow Analysis, Municipal Solid Waste, Circular Economy, Resource Recovery, Sustainability Assessment, Waste Management

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List of Abbreviations

MSW Municipal Solid Waste

GHG Greenhouse Gas

MFA Material Flow Analysis

CEV Centre d'Enfouissement et de Valorisation

Note:

A list of abbreviations does not release the user from the obligation to explain an abbreviation when using it for the first time by stating the long form.

Only subject-specific abbreviations must be included in the list of abbreviations. Common abbreviations such as "e.g., i.e., cf, etc". may be omitted.

Introduction

As urban areas grow rapidly, municipal solid waste (MSW) management has become a major challenge due to increasing waste generation, limited landfill capacity, and rising environmental concerns. The conventional waste management system heavily relies on landfilling, which has resulted in inefficient resource use, increased greenhouse gas (GHG) emissions, and lost opportunities for material recovery.

To transition toward sustainable waste management, the Circular Economy approach is necessary, whereby waste is considered a resource instead of a disposal issue. Environmental impacts can be reduced and overall system efficiency improved by implementing strategies such as source segregation, recycling, composting, and resource recovery.

This study analyzes different municipal solid waste management strategies for the city of Marrakech using Material Flow Analysis (MFA) and Cost Efficiency Analysis to assess how various waste treatment pathways impact landfill dependency, recycling performance, resource recovery potential, and economic feasibility while incorporating environmental and economic indicators to identify sustainable waste management strategies that align with long-term environmental and resource management goals.

Objective

The primary objectives of this study are:

- Quantify municipal solid waste flows under different management scenarios.
- Evaluate recycling, composting, and landfill diversion pathways using Material Flow Analysis (MFA).
- Assess resource recovery opportunities and Circular Economy potential.
- Compare the environmental performance of alternative waste management systems.
- Evaluate the economic implications of different collection and treatment approaches.
- Identify sustainable waste management strategies that balance environmental benefits and operational costs.

Methodology

Material flow analysis (MFA) was used to estimate the movement of waste materials through various treatment and disposal pathways. MFA is a method to systematically assess the material inputs, outputs, stocks, and flows of a defined system.

Three waste management scenarios were developed and compared:

a) Scenario 1: Baseline Landfill-Oriented System
   The first scenario is the current waste management system, which involves minimal sorting of waste, and most of the municipal solid waste (MSW) is sent to landfill facilities, with recycling activities largely reliant on informal recovery systems.
b) Scenario 2: Improved Recycling System
   In the second scenario, there is better sorting and material recovery, and fewer recyclable materials go to landfills.
c) Scenario 3: Integrated Circular Economy System
   In the third scenario, the waste management strategy is based on source segregation, recycling, and composting, which is a Circular Economy-oriented waste management strategy designed to maximize resource recovery and minimize landfill dependency.

Study Area and System Definition

The case study looks at the municipal solid waste management system in Marrakech, Morocco, which has an estimated population of 1,033,050 people in 2022 (assuming the exercise assumptions, the solid waste generation rate in the city is 0.76 kg per capita per day, and the formal waste collection rate is 100%, i.e., all generated waste enters the formal collection system).

The waste management system is made up of formal waste collection companies, transfer stations, informal waste pickers, and the Centre d'Enfouissement et de Valorisation (CEV), which is both a sorting facility and a controlled landfill site. At the transfer stations, recyclable materials such as plastics, metals, and glass are recovered by informal waste pickers before the waste is transported to the CEV.

Material Flow Analysis Modelling Approach

A material flow analysis (MFA) was performed for MSW using the STAN (SubSTance Flow ANalysis) software package, which is a tool for the quantification and visualization of material flows in defined systems based on the principle of mass conservation (i.e., material inputs and outputs and stock changes are balanced within a system boundary).

The MFA model evaluated the flows of MSW through collection, transfer stations, sorting facilities, composting units, recycling pathways, and final landfill disposal. To analyze the impact of increased material recovery, biological treatment, and source segregation on the overall system performance, three alternative waste management scenarios were modeled based on key waste fractions such as organic waste, paper, plastics, glass, metals, and residual waste, and scenario modeling was conducted to compare the landfill disposal, recycling rates, compost production, and resource recovery performance.

Cost Efficiency Analysis

The Material Flow Analysis and Cost Efficiency Analysis were created using different datasets used in course exercises. The cost assessment is based on municipal waste generation data at the district level (approximately 69,530 tonnes per year), while the MFA exercise uses a system-wide estimate of waste generation based on population (286,568 tonnes per year) and, as a result, reported waste quantities are not directly comparable and should be interpreted within the context of the specific assessment frameworks they belong to. The two datasets are not directly combined in this study, but are evaluated separately.

Figure 1: **Integrated Assessment Framework (Author)**

Waste Characterization

Characterization of waste streams is essential to understanding the generation and composition of municipal solid waste and plays a crucial role in assessing treatment technologies, resource recovery opportunities, and landfill diversion strategies. The waste stream analyzed in this study includes organic waste, paper, plastics, glass, metals, textiles, and miscellaneous residual waste fractions. Organic waste is the largest fraction of the waste stream and can be composted or treated biologically, while recyclable fractions like paper, plastics, glass, and metals have potential for material recovery and Circular economy applications.

Waste Type	**Total Waste (tons)**
Organic	35,953.11
Paper	12,305.15

Plastic	7,214.74
Glass	3193.18
Metals	2,753.41
Others (including textiles)	13,561.74

Table 1: Composition of Municipal Solid Waste Generated within the Study Area

District	Income Level	Waste Value (in Ton)
1	High	9,085.83
2	Upper-middle	2,608.78
3	High	18,171.83
4	Low	1,202.44
5	Low	1,275.68
6	Low	2,004
7	Lower-middle	13,949.57
8	Lower-middle	7,121.68
9	Upper-middle	14,109.91

Table 2: Annual Municipal Solid Waste Generation by District

The analysis indicates that a substantial proportion of the waste stream could be diverted from landfill through improved source segregation, recycling infrastructure, and organic waste treatment systems. These findings form the basis for the development of alternative waste management scenarios evaluated in this study.

Scenario Development

Three waste management scenarios were developed based on the assumptions provided in the MFA exercise. The scenarios progressively increase the complexity of material recovery and treatment.

a) Scenario 1: Semi-Automated Sorting and Landfilling

All municipal solid waste is transported to the CEV, where semi-automated sorting recovers plastics, metals, and glass, which are sold to recycling facilities. In contrast, the remaining waste is disposed of in a controlled landfill.

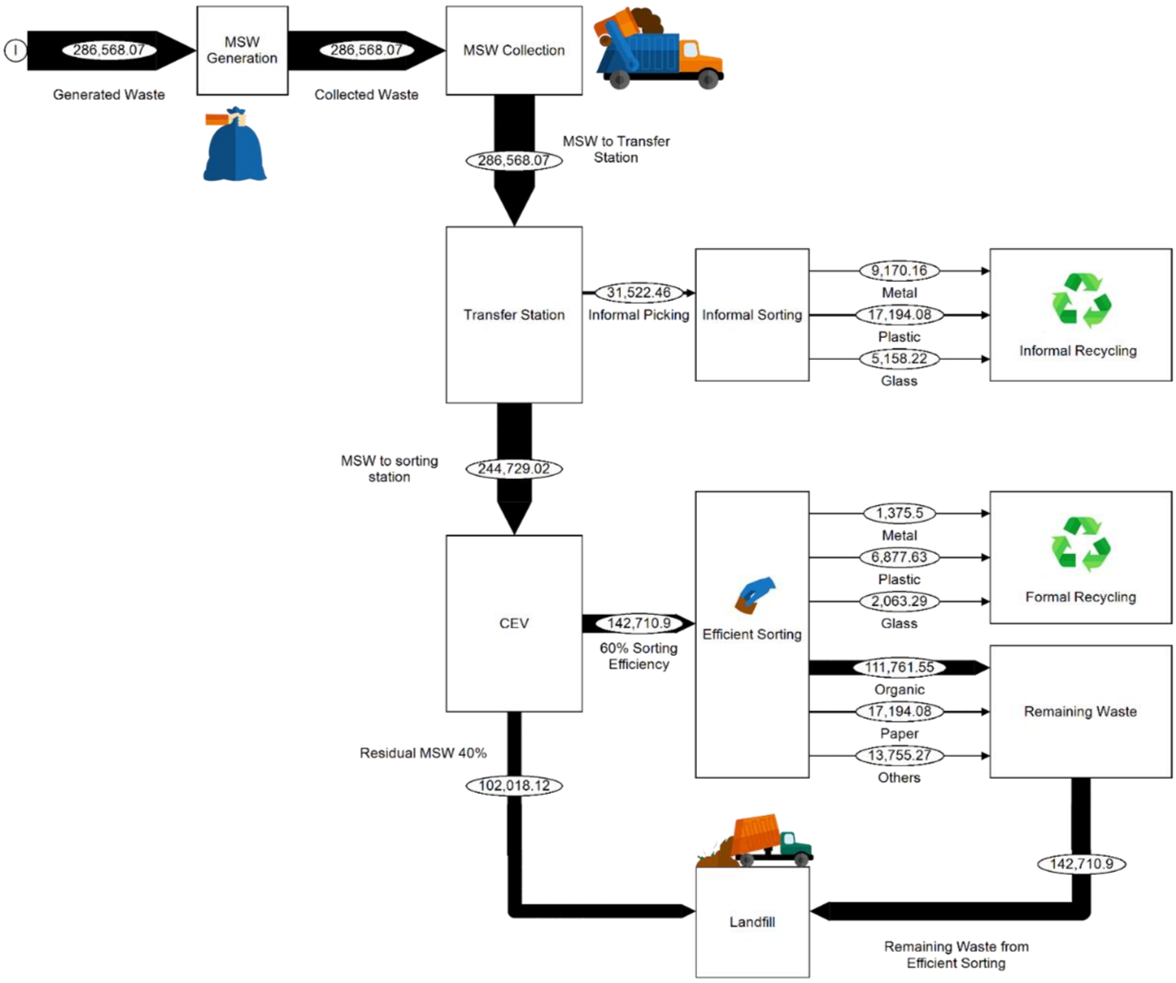

Figure 2: STAN-based material flow model for Scenario 1 (semi-automated sorting and landfill disposal)

Scenario I is the baseline waste management configuration and reflects a disposal-based system with a low level of resource recovery, limited material recovery (separated through semi-automated sorting processes at the CEV facility), high landfill dependency, and no biological treatment or source segregation, which is used as a reference case to evaluate environmental improvements in the alternative scenarios.

b) Scenario 2: Sorting and Biological Stabilization

The sorting processes from Scenario I are retained, and a biological stabilization and composting unit is introduced to treat organic fractions, reducing waste mass by about 50 percent through decomposition.

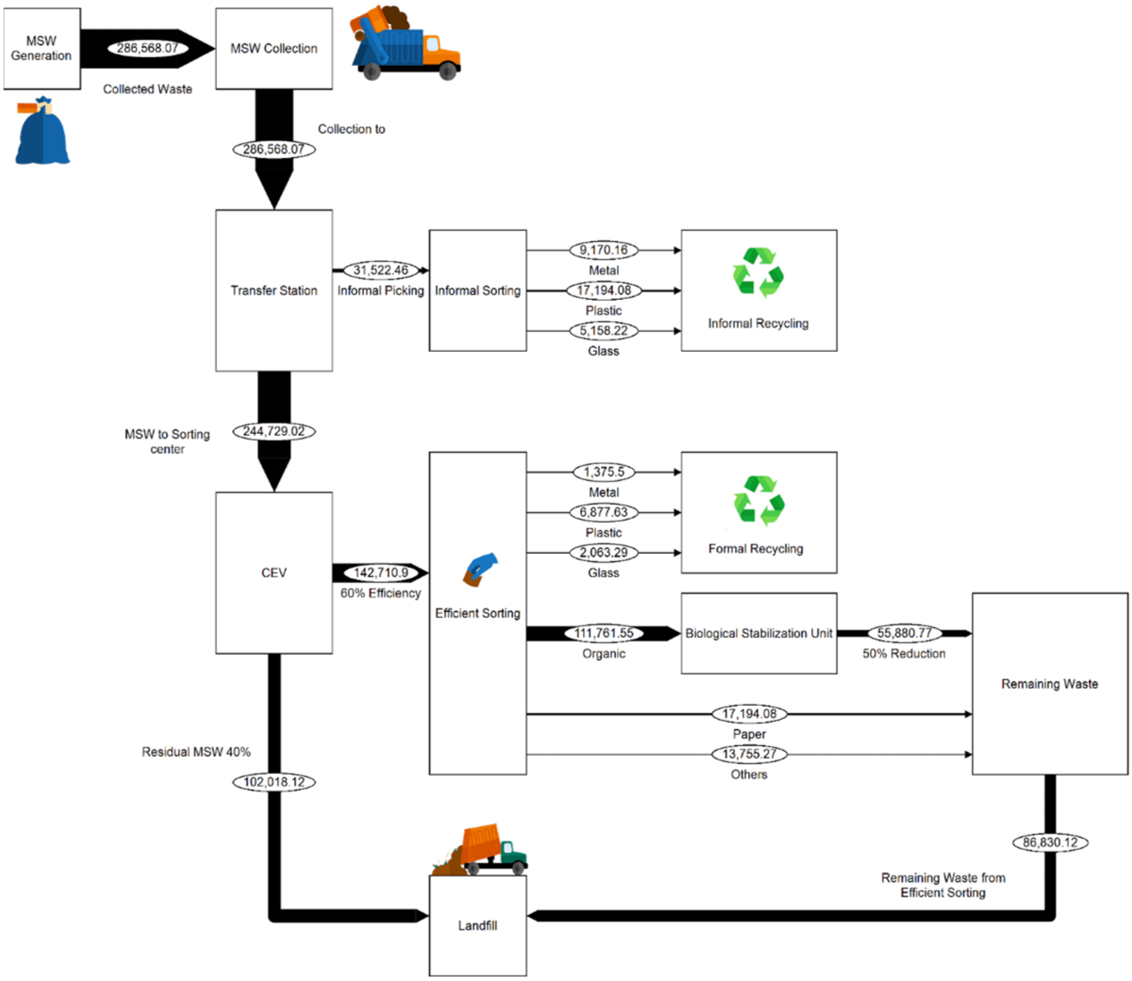

Figure 3: STAN-based material flow model for Scenario 2 (sorting and biological stabilization)

Compared to the baseline system, this scenario increases resource recovery rates and improves overall material utilization while reducing dependence on final disposal facilities.

c) Scenario 3: Source Segregation and Composting

Source segregation of organic waste is introduced, with about 70 percent of the organic waste collected directly from the source and sent to a composting facility, while residual waste follows the same collection and sorting path as in Scenario I. The compost can be used for landscaping and gardening.

This scenario aims to maximize resource recovery, minimize reliance on landfills, and improve environmental performance across the entire waste management system.

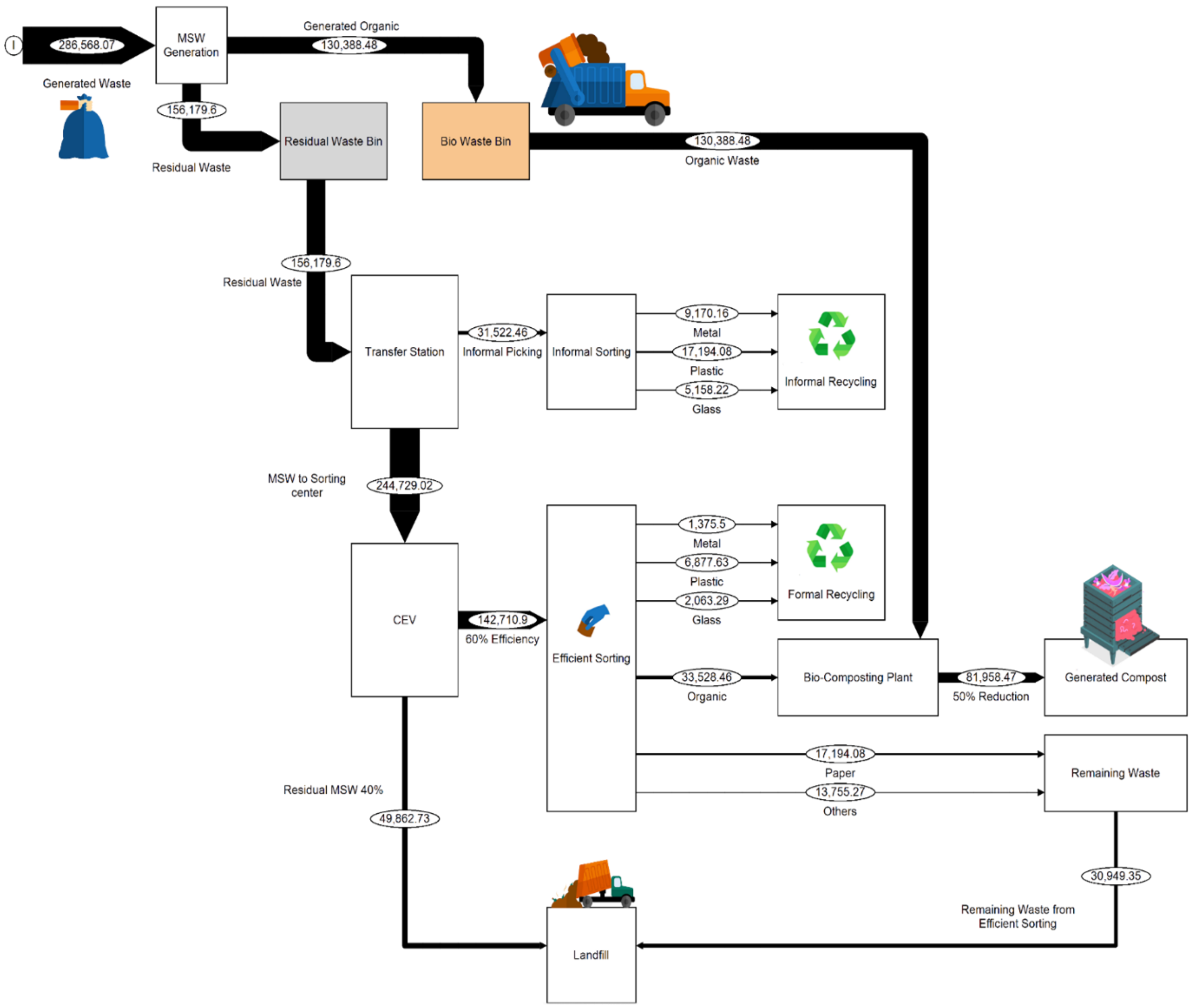

Figure 4: STAN-based material flow model for Scenario 3 (source segregation, recycling, and composting)

Results

Figure 5: Summary of environmental and economic performance across alternative municipal solid waste management scenarios (Author)

Environmental Performance Assessment

The MFA shows that the three scenarios' resource recovery performance differs significantly. Scenario 1 exhibits the highest reliance on landfill disposal, resulting in significant losses of potentially recoverable materials. Opportunities for resource recovery and circular material flows are limited by low recycling activity.

Scenario 2 uses improved material recovery procedures to increase recycling performance. Increasing the diversion of recyclable fractions reduces demand for landfills and boosts resource recovery.

The best environmental performance is obtained in Scenario 3. Composting and source segregation maximize the recovery of valuable material resources while drastically lowering the need for landfill disposal.

Indicator	Scenario 1	Scenario 2	Scenario 3
Total MSW (t/year)	286,568	286,568	286,568
Recycled Waste (t/year)	41,839	41,839	41,839
Landfilled Waste (t/year)	244,729	188,848	163,917
Compost Produced (t/year)	0	0	80,812

Table 3: Comparison of waste treatment outcomes and landfill diversion across the three waste management scenarios

Scenario 3 (with source segregation, recycling, and composting pathways) achieved the best environmental performance; landfill disposal was reduced by 80,812 tonnes/year (33% landfill diversion compared to Scenario I), and diversion of organic waste to composting facilities greatly reduced the amount of residual waste needing final disposal and produced valuable compost products for landscaping and agricultural uses.

In addition to landfill diversion, Scenario 3 demonstrated the strongest resource recovery performance. The recovery of recyclable materials combined with organic waste treatment increased the overall utilization of waste-derived resources and supported the transition from a disposal-oriented system towards a Circular Economy model.

Cost-Efficiency Assessment

A cost efficiency assessment was conducted comparing the economic performance of two municipal solid waste management systems in Marrakech: Scenario 1 (conventional Landfill-Oriented collection system) and Scenario 2 (decentralized collection system with resource recovery activities) for an annual average waste generation of approximately 69,530 tonnes per year in nine districts.

Scenario	Annual Cost (US$)
Scenario 1 – Conventional Collection & Landfill Disposal	3,476,494
Scenario 2 – Decentralized Collection & Resource Recovery	3,984,667

Table 4: Annual operational costs associated with alternative waste management systems

Scenario 2 had annual operating costs approximately **US$508,000** higher than the conventional system (primarily due to additional collection, segregation, composting, and resource recovery activities), but it demonstrated improved environmental performance through increased resource recovery and reduced reliance on landfill. The findings indicate that short-term economic costs are traded off against long-term sustainability benefits when environmental considerations are integrated into municipal waste management. These additional investments support the transition towards more resource-efficient and circular waste management systems.

Resource Recovery Assessment

The findings show that overall, resource recovery performance increases from Scenario 1 to Scenario 3, with Scenario 3 having the highest overall recovery potential through the combination of formal recycling systems with source-separated organic waste treatment, maximizing the value retained within the waste management system, reducing the need for virgin material extraction, and aligning with Circular Economy goals of reusing recovered materials in productive cycles.

Discussion

A comprehensive framework for evaluating municipal waste management systems from both environmental and economic perspectives is to integrate material flow analysis and cost efficiency analysis. The results show that landfill-dependent waste management systems lose the most recoverable materials and generate the most environmental burdens, even though they have lower short-term operational costs.

Recycling and composting support the move towards the circular economy and keep material value in the economy, while reducing virgin resource extraction. Source segregation is found to be a key enabler for enhancing material quality and maximizing recovery rates.

This comparison of scenarios illustrates that environmental performance and economic efficiency cannot be evaluated in isolation. Balancing the operational costs with resource recovery benefits, landfill diversion objectives, and long-term environmental outcomes will help guide sustainable waste management.

Results also show that municipal decision-makers should apply integrated assessment approaches when considering alternative waste management strategies by combining environmental and economic indicators to make more informed planning decisions and build more robust urban waste systems.

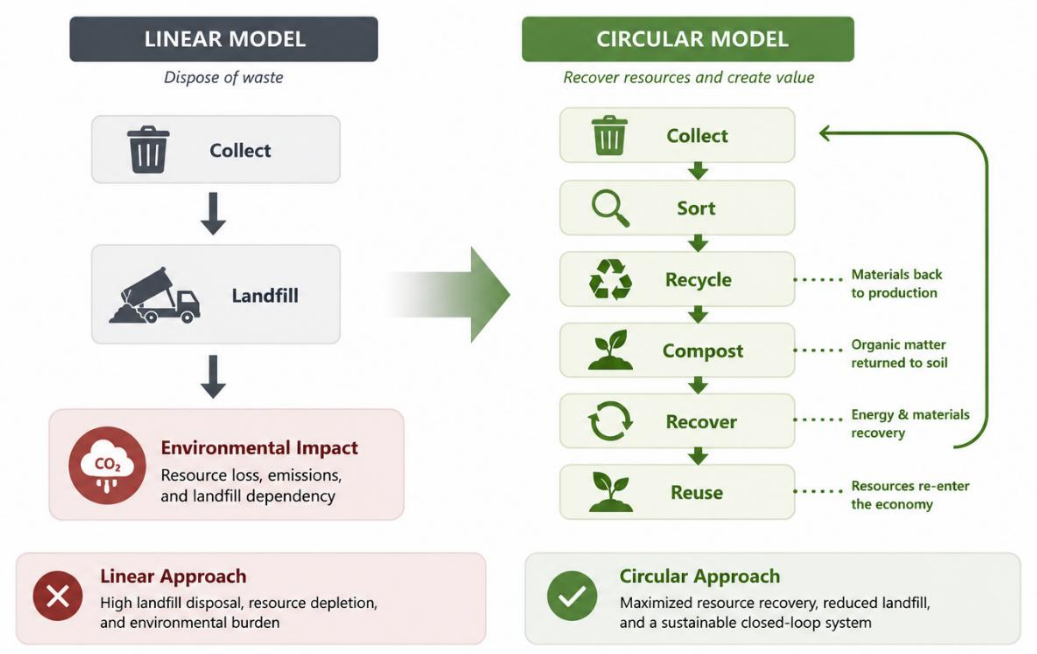

Figure 6: Transition from Linear to Circular Waste Management (Author)

Recommendations

The following recommendations can be made based on the results of this study:

a) Implement source segregation programs to enhance the quality and number of recyclable materials.
b) Expand recycling infrastructure and formal collection systems to enhance material recovery rates and reduce landfill dependency.
c) Establish composting facilities to manage organic waste, given the high organic content of the municipal waste stream.
d) Reduce landfill dependency through the gradual implementation of circular economy approaches.
e) Establish monitoring systems for indicators of waste generation, recycling performance, and resource recovery.
f) Include environmental performance metrics in municipal waste management planning and investment decisions.
g) Engage public awareness and stakeholder participation to enhance waste separation practices and support sustainable waste management initiatives.

Conclusion

This research compared different municipal solid waste management systems in Marrakech by integrating Material Flow Analysis and Cost Efficiency Analysis. The findings suggest that recycling, composting, and source segregation-based waste management strategies are more effective in improving resource recovery and environmental performance than the conventional Landfill-Oriented systems. However, these strategies involve extra investment and operational costs with substantial benefits from landfill diversion, material utilization, and environmental impact reduction.

The results corroborate that the circular economy principles can help to turn municipal waste management systems from disposal-oriented systems into resource recovery-oriented systems. MFA and Cost Efficiency Analysis can be used as decision-support tools for municipal waste management to enhance sustainability performance and waste management objectives in the long-term.

The study demonstrates how integrated environmental and economic assessments can support evidence-based decision-making for sustainable urban waste management. Additional sustainability indicators such as Life Cycle Assessment (LCA), greenhouse gas (GHG) quantification, and social impact indicators may also be considered in future research.

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